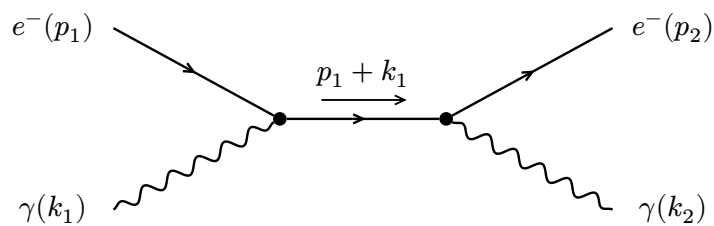


CeTZ Feynman Lite

Feynman diagrams with Typst and CeTZ



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Development edition · September 2026

Examples verified with Typst 0.15.0 and CeTZ 0.5.2. The API reference is generated with tidy 0.4.3.

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Using this guide

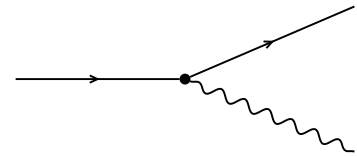
Examples use the versioned package import. Before publication, install the local release candidate under a Typst package path to use these examples. The library draws diagrams; it does not calculate amplitudes or validate physical processes.

1 Your first diagram

Call `feynman` inside a CeTZ canvas. Each edge specifies its start ID, end ID and options. Vertices are inferred from these IDs; `incoming` and `outgoing` identify the external legs.

```
#import "@preview/cetz:0.5.2"
#import "@preview/cetz-feynman-lite:0.1.0": feynman

#ceztz.canvas(length: 10mm, {
  feynman(
    ("i", "v", (particle: "fermion")),
    ("v", "o", (particle: "fermion")),
    ("v", "g", (particle: "photon")),
    incoming: ("i",),
    outgoing: ("o", "g"),
  )
})
```

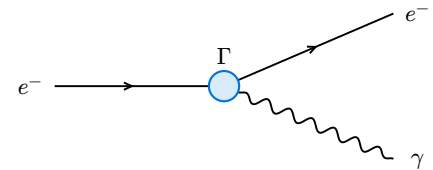


2 Customizing vertices

Add `vertices` to set markers and labels. Coordinates remain optional. With explicit vertices, set each vertex's `role` instead of passing top-level `incoming` or `outgoing`. A blob is circular, and `size` is its radius.

```
#import "@preview/cetz:0.5.2"
#import "@preview/cetz-feynman-lite:0.1.0": feynman

#cezt.canvas(length: 10mm, {
  feynman(
    ("i", "v", (particle: "fermion")),
    ("v", "o", (particle: "fermion")),
    ("v", "g", (particle: "photon")),
    vertices: (
      (id: "i", role: "incoming", label: $e^-$,
        label-offset: (-0.4, 0)),
      (id: "v", marker: "blob", size: 0.25,
        fill: blue.lighten(85%),
        stroke: 1pt + blue,
        label: $Gamma$, label-offset: (0, 0.5)),
      (id: "o", role: "outgoing", label: $e^-$,
        label-offset: (0.4, 0)),
      (id: "g", role: "outgoing", label: $gamma$,
        label-offset: (0.4, 0)),
    ),
  ),
})
```

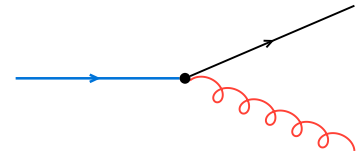


3 Propagator styles

`particle` selects a preset; `style` overrides its appearance. Line thickness uses physical lengths such as pt. Amplitude, wavelength and offsets use canvas units, whose physical size is set by `length`. `gluon-aspect` controls the longitudinal excursion of the coil.

```
#import "@preview/cetz:0.5.2"
#import "@preview/cetz-feynman-lite:0.1.0": feynman

#ceztz.canvas(length: 10mm, {
  feynman(
    ("i", "v", (particle: "fermion",
      style: (paint: blue, thickness: 1.2pt))),
    ("v", "o", (particle: "fermion")),
    ("v", "g", (particle: "gluon",
      style: (paint: red, amplitude: 0.16,
        wavelength: 0.45, gluon-aspect: 1.1))),
    incoming: ("i",),
    outgoing: ("o", "g"),
  )
})
```

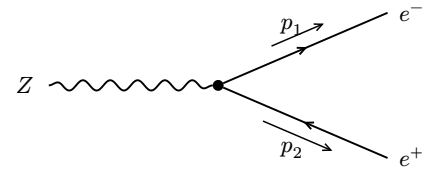


4 Fermion flow and momentum

In $Z \rightarrow e^- e^+$, the positron's fermion-flow arrow points toward the vertex, while its momentum points outward. Set `arrow` and `momentum` independently. Directions and sides are relative to the edge's start $\rightarrow$ end order.

```
#import "@preview/cetz:0.5.2"
#import "@preview/cetz-feynman-lite:0.1.0": feynman

#ceztz.canvas(length: 10mm, {
  feynman(
    ("i", "v", (particle: "photon")),
    ("v", "o", (particle: "fermion",
      momentum: (label: $p_1$, side: "left"))),
    ("v", "j", (particle: "fermion",
      arrow: "backward",
      momentum: (label: $p_2$,
        direction: "forward",
        side: "right", span: 0.4))),
    vertices: (
      (id: "i", role: "incoming", label: $Z$,
        label-offset: (-0.4, 0)),
      (id: "v"),
      (id: "o", role: "outgoing", label: $e^-$,
        label-offset: (0.4, 0)),
      (id: "j", role: "outgoing", label: $e^+$,
        label-offset: (0.4, 0)),
    ),
  ),
})
```

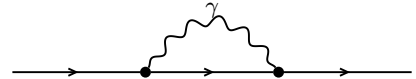


5 Parallel edges and layout

In this electron self-energy diagram, the electron and photon connect the same pair of vertices. `main-lines` aligns the electron line, and `route` places the photon above it. No vertex coordinates are required. An edge cannot combine `route` with `bend` or `controls`.

```
#import "@preview/cetz:0.5.2"
#import "@preview/cetz-feynman-lite:0.1.0": feynman

#ceztz.canvas(length: 10mm, {
  feynman(
    ("i", "a", (particle: "fermion")),
    ("a", "b", (particle: "fermion")),
    ("b", "o", (particle: "fermion")),
    ("a", "b", (particle: "photon",
      label:  $\gamma$ ,
      route: (side: "above", level: 1))),
    incoming: ("i",), outgoing: ("o",),
    layout: (main-lines: (("i", "a", "b",
      "o"),)),
  )
})
```

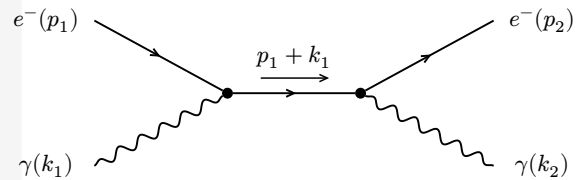


6 Compton scattering: s channel

$e^- \gamma \rightarrow e^- \gamma$: incoming particles are on the left, outgoing particles on the right. Following the electron line, absorption precedes emission. The internal momentum is $q_s = p_1 + k_1$, with $s = q_s^2$. The tree-level amplitude is $\mathcal{M} = \mathcal{M}_s + \mathcal{M}_u$.

```
#import "@preview/cetz:0.5.2"
#import "@preview/cetz-feynman-lite:0.1.0": feynman

#ceztz.canvas(length: 10mm, {
  feynman(
    ("i", "a", (particle: "fermion")),
    ("j", "a", (particle: "photon")),
    ("a", "b", (particle: "fermion",
      momentum: (label:  $p_1 + k_1$ ,
        label-offset: 0.35,
        span: 0.5))),
    ("b", "o", (particle: "fermion")),
    ("b", "g", (particle: "photon")),
    vertices: (
      (id: "i", role: "incoming",
        label:  $e^-(p_1)$ ,
        label-offset: (-0.8, 0)),
      (id: "j", role: "incoming",
        label:  $\gamma(k_1)$ ,
        label-offset: (-0.8, 0)),
      (id: "a"), (id: "b"),
      (id: "o", role: "outgoing",
        label:  $e^-(p_2)$ ,
        label-offset: (0.8, 0)),
      (id: "g", role: "outgoing",
        label:  $\gamma(k_2)$ ,
        label-offset: (0.8, 0)),
    ),
    layout: (main-lines: (("a", "b")),
      incoming-order: ("i", "j"),
      outgoing-order: ("o", "g")),
  )
})
```

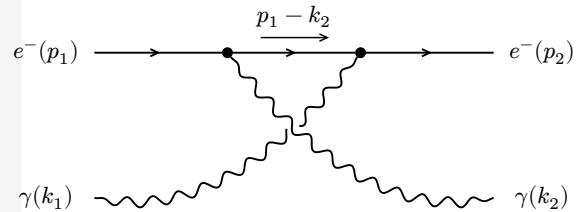


7 Compton scattering: u channel

Following the electron line, emission precedes absorption. The internal momentum is $q_u = p_1 - k_2 = p_2 - k_1$, with $u = q_u^2$. The crossing of the two photon lines is not a vertex; **crossing-gap** opens a gap in the lower line.

```
#import "@preview/cetz:0.5.2"
#import "@preview/cetz-feynman-lite:0.1.0": feynman

#ceztz.canvas(length: 10mm, {
  feynman(
    ("i", "a", (particle: "fermion")),
    ("j", "b", (particle: "photon",
      route: (side: "below", level: 1))),
    ("a", "b", (particle: "fermion",
      momentum: (label:  $p_1 - k_2$ ,
        label-offset: 0.35,
        span: 0.5))),
    ("b", "o", (particle: "fermion")),
    ("a", "g", (particle: "photon",
      route: (side: "below", level: 1))),
    vertices: (
      (id: "i", role: "incoming",
        label:  $e^-(p_1)$ , label-offset: (-0.8, 0)),
      (id: "j", role: "incoming",
        label:  $\gamma(k_1)$ , label-offset: (-0.8, 0)),
      (id: "a",),
      (id: "b",),
      (id: "o", role: "outgoing",
        label:  $e^-(p_2)$ , label-offset: (0.8, 0)),
      (id: "g", role: "outgoing",
        label:  $\gamma(k_2)$ , label-offset: (0.8, 0)),
    ),
    layout: (main-lines: (("i", "a", "b", "o")),),
    incoming-order: ("i", "j"),
    outgoing-order: ("o", "g"),
    crossing-gap: 0.12,
  )
})
```



8 API reference

Import the public API from `lib.typ`. Coordinates, radii, amplitudes, wavelengths and offsets use canvas units. CeTZ's `length` sets their physical scale. Line widths use Typst lengths; labels accept Typst content.

8.1 Diagram functions

- `diagram-data()`
- `feynman()`

diagram-data

Normalize, lay out and sample a diagram. Return vertices, edges, positions and routes without drawing. With explicit vertices, specify external roles on each vertex.

Parameters

```
diagram-data(
  ..connections: array dictionary ,
  vertices: array ,
  edges: array ,
  incoming: array ,
  outgoing: array ,
  registry: dictionary ,
  layout: dictionary
) -> dictionary
```

..connections array or dictionary

Edge tuples or dictionaries. Cannot be combined with non-empty edges.

vertices array

Vertex dictionaries. If omitted, infer vertices from edge endpoints.

Default: ()

edges array

An array of edges; tuples and dictionaries may be mixed.

Default: ()

incoming array

Incoming external IDs when vertices are inferred.

Default: ()

outgoing array

Outgoing external IDs when vertices are inferred. Must not overlap incoming.

Default: ()

registry `dictionary`

A mapping from particle names to propagator styles.

Default: `styles`

layout `dictionary`

Optional main lines, external ordering and spacing. See the layout fields.

Default: `(:)`

feynman

Call inside `cetz.canvas` to return CeTZ drawing elements.

Parameters

```
feynman(
  ..args: any,
  crossing-gap: int float
) -> array
```

..args `any`

Accepts all diagram-data arguments, including edges, vertices, registry and layout.

crossing-gap `int` or `float`

Crossing gap half-length in canvas units. Zero disables gaps; only arrowless wave/wave crossings are handled.

Default: `0`

8.2 Rendering and registration

- `draw-diagram()`
- `render()`

draw-diagram

Draw an existing diagram-data result as CeTZ drawing elements.

Parameters

```
draw-diagram(
  result: dictionary,
  crossing-gap: int float
) -> array
```

result dictionary

The dictionary returned by diagram-data.

crossing-gap int or float

Crossing gap half-length in canvas units; zero disables gaps.

Default: 0

render

Render existing diagram data in a standalone CeTZ canvas.

Parameters

```
render(
  result: dictionary,
  unit: length,
  crossing-gap: int float
) -> content
```

result dictionary

The dictionary returned by diagram-data.

unit length

The physical length of one canvas unit.

Default: 9mm

crossing-gap int or float

Crossing gap half-length in canvas units; zero disables gaps.

Default: 0

- `register-style()`

Variables

- `styles`

`register-style`

Return a registry with a new preset. The original is unchanged; duplicate names are rejected.

Parameters

```
register-style(
  registry: dictionary,
  name: str,
  definition: dictionary
) -> dictionary
```

registry `dictionary`

The original propagator registry.

name `str`

A non-empty name for the new preset.

definition `dictionary`

A style dictionary; see the style fields. Custom rendering callbacks are not supported.

styles `dictionary`

Built-in propagator presets: scalar, fermion, photon, gluon, ghost and double.

8.3 Vertices

```
(id: "v", marker: "blob", size: 0.25,
  fill: blue.lighten(85%), stroke: 1pt + blue,
  label:  $\gamma$ , label-offset: (0, 0.45))
```

Field	Default / type	Description
id	Required / str	A unique, non-empty vertex ID.
role	"interaction"	interaction, incoming or outgoing. Sets the default marker and layout role, not arrow direction.
at	auto	A fixed (x, y) coordinate, or auto. Both coordinates must be finite numbers.
marker	By role / str	none, dot, cross, circle or blob. Defaults to dot for interactions and none for external legs.
size	0.09 / positive number	Circle radius, or the horizontal and vertical half-size of a cross.
fill	black	Fill for dot and blob. A circle is always transparent.
stroke	0.9pt + black	CeTZ stroke for circle, blob and cross. A dot has no outline.
label	none / content	Vertex label, including mathematical or formatted content.
label-offset	(0, 0.3)	Label displacement relative to the vertex center.

The dot, circle and blob markers are circular. Propagator centerlines are trimmed to their boundaries, including both ends of a self-loop. A cross connects at its center. The none marker hides the symbol while retaining its topology. Custom marker callbacks and vertex registries are not available.

8.4 Edges and paths

Both forms below are accepted and may be mixed in an edge array.

```
("a", "b", (particle: "fermion", label: $e^-$))
(id: "electron", from: "a", to: "b", particle: "fermion")
```

Field	Default / type	Description
id	Generated / str	A unique, non-empty ID. Explicit IDs stabilize parallel-edge allocation and crossing order.
from / to	Required / str	Endpoint IDs. Equal IDs create a self-loop. Tuple options cannot override endpoints.
particle	“scalar” / str	A preset name in registry. Names are drawing conventions, not physics validation.
style	(:) / dictionary	Local overrides of the selected preset.
arrow	From style / str	none, forward or backward relative to from $\rightarrow$ to. Overrides style.arrow.
bend	auto	A number from -1 to 1 . Zero is straight; ± 1 is a semicircle. Positive bends to the left of from $\rightarrow$ to.
controls	none	Two absolute control points for a cubic Bézier. Not for self-loops; cannot combine with explicit bend or route.
route	none / dictionary	side: above or below (default above); level: positive integer (default 1). Uses absolute canvas directions.
loop-size	0.65 / positive number	Self-loop half-axis in its facing direction. Self-loops only.
loop-aspect	1 / positive number	Transverse half-axis divided by the facing half-axis. Self-loops only.
loop-angle	auto / angle	Self-loop direction: 0deg right, 90deg up.
label	none / content	Propagator label, independent of momentum.label.
label-at	0.5 / number	Normalized arc-length position on the complete centerline, from 0 to 1.
label-offset	0.22 / number	Offset along the left normal; negative values move to the right.
momentum	none / dictionary	Independent momentum annotation; see Momentum.

A route requires horizontal separation and cannot be used on a self-loop. Its midpoint is displaced by $\pm 0.85 \times$ level relative to the endpoint midpoint. The level controls height, not crossing order. Explicit paths through unrelated vertices produce an error rather than being silently rerouted.

8.5 Styles

Field	Default / type	Description
line	“solid” / str	solid, dashed, dotted, wave, coil or double. Presets override this base default.
arrow	“none” / str	none, forward or backward.
paint	black	Color of the propagator and its arrows.
thickness	0.9pt / positive length	Line width of the propagator and its arrows.
amplitude	0.085 / positive number	Wave or coil amplitude; half-spacing for double. The gluon preset uses 0.168.
wavelength	0.42 / positive number	Target wave or coil pitch, adjusted to fit the path.
gluon-aspect	1.1 / non-negative number	Longitudinal coil excursion relative to amplitude. Applies to coil only.

Preset	Line	Arrow
scalar	dashed	none
fermion	solid	forward
photon	wave	none
gluon	coil	none
ghost	dotted	forward
double	double	none

Precedence: base defaults → registry preset → edge style → edge arrow. All numerical style values must be finite. Waves taper at the endpoints; coils fit whole turns plus a final half turn, scaling down for short edges. Double lines converge at their endpoints.

```
#let custom = register-style(styles, "W", (
  line: "wave", paint: blue,
))
// Pass registry: custom to feynman.
// Use particle: "W" on an edge.
```

Registration combines existing line styles; it does not add a rendering algorithm.

8.6 Momentum

```
momentum: (
  label:  $p$ , direction: "forward", side: "left",
  at: 0.5, span: 0.3, offset: 0.25, label-offset: 0.35,
)
```

Field	Default / type	Description
label	none / content	Momentum label. An empty dictionary draws an unlabeled arrow; momentum: none disables the annotation.
direction	"forward" / str	forward or backward relative to from $\rightarrow$ to, independent of the propagator arrow.
side	"left" / str	left or right relative to from $\rightarrow$ to. Reversing direction does not change the side.
at	0.5 / number	Midpoint of the arrow interval, in normalized arc length of the complete centerline.
span	0.3 / positive number	Fraction of the path covered by the arrow. Both at $-$ span/2 and at $+$ span/2 must lie between 0 and 1.
offset	0.25 / positive number	Distance from the centerline to the momentum arrow in canvas units. The label moves with the arrow.
label-offset	0.20 / non-negative number	Additional outward displacement of the label from the momentum arrow, in canvas units. Must be finite; zero places the label on the arrow path.

Use momentum.label-offset to move only the label, leaving the arrow unchanged. The label is placed at at, outward on the side selected by momentum.side. For example, label-offset: 0.35 gives more clearance than the default 0.20. The edge's label-offset affects only the propagator label.

The fermion-flow arrow sits at the midpoint of the trimmed visible path. Its position, head shape and dimensions are not exposed as options. Both arrow types use the propagator color and line width.

8.7 Layout and crossings

Field	Default / type	Description
main-lines	() / array	Under layout: arrays of vertex IDs. Each line runs left to right; lines run top to bottom. Each requires at least two adjacent connected vertices. No repeated or shared vertices.
incoming-order	() / array	Under layout: top-to-bottom incoming order. If non-empty, list every incoming vertex exactly once.
outgoing-order	() / array	Under layout: top-to-bottom outgoing order. If non-empty, list every outgoing vertex exactly once.
external-gap	2.4 / positive number	Under layout: vertical spacing of automatically placed external legs. Fixed and main-line legs stay in place; spacing between fixed anchors may differ.
crossing-gap	0 / non-negative number	Drawing option on feynman, draw-diagram and render. Gap half-length along the wave. Zero disables gaps. Only arrowless wave/wave crossings are handled.

On a main line, an incoming vertex may appear only first, and an outgoing vertex only last. Fixed coordinates must satisfy horizontal alignment and ordering. Candidate edges between adjacent main-line vertices are selected by ID for the default straight path; explicit paths take precedence.

At a crossing, the lexicographically larger edge ID stays continuous. Shared endpoints, regions near vertices and waves with flow arrows are excluded.

8.8 Returned data

Field	Default / type	Description
vertices	array	Vertex dictionaries with defaults filled in.
edges	array	Normalized edges with resolved style and arrow values.
positions	dictionary	Vertex ID $\rightarrow$ final (x, y) position.
routes	array	Edge properties plus path, table, start and end; pass the result to draw-diagram or render.

path describes the centerline. table stores sampled points, cumulative lengths and total length. start and end delimit the trimmed visible arc-length interval. Label positions and momentum intervals refer to the complete centerline.

Unknown fields, duplicate IDs, invalid values and unsatisfiable constraints are rejected. Layout and collision checks use finite rules and sampling. There is no automatic label or momentum avoidance, nor validation of charge, momentum conservation or amplitudes.